

Exercise of Supervised Learning: Advanced Risk Minimization Part 2

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Exercise 1: Risk Minimizers for 0-1-Loss

Setting. Classification with $\mathcal{Y} = \{1, \dots, g\}$ and $\mathcal{H} = \{h : \mathcal{X} \rightarrow \mathcal{Y}\}$. The loss of interest is the 0-1-loss:

$$L(y, h(\mathbf{x})) = I_{y \neq h(\mathbf{x})} = \begin{cases} 1, & \text{if } y \neq h(\mathbf{x}), \\ 0, & \text{if } y = h(\mathbf{x}). \end{cases}$$

Question (a). Take the hypothesis space of *constant models*

$$\mathcal{H} = \{h \mid h(\mathbf{x}) = \theta \in \mathcal{Y} \forall \mathbf{x} \in \mathcal{X}\}.$$

Show that

$$\hat{h}(\mathbf{x}) = \text{mode} \left\{ \mathcal{Y}^{(i)} \right\}$$

is the empirical risk minimizer for the 0-1-loss.

Roadmap (general recipe for ERM w/ const. models)

1. Write down the empirical risk $\mathcal{R}_{\text{emp}}(h)$.
2. Substitute $h(\mathbf{x}) = \theta$ (hypothesis is constant).
3. Check strict convexity in θ .
(not needed here)
4. Set the gradient to 0 and solve for θ^* .
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Solution (a) — Steps 1 & 2: write $\mathcal{R}_{\text{emp}}(h)$, then substitute $h(\mathbf{x}) = \theta$

Recall the 0-1-loss:

$$L(y, h(\mathbf{x})) = I_{y \neq h(\mathbf{x})} = \begin{cases} 1, & \text{if } y \neq h(\mathbf{x}), \\ 0, & \text{if } y = h(\mathbf{x}). \end{cases}$$

Step 1 — Write down the empirical risk.

$$\mathcal{R}_{\text{emp}}(h) = \sum_{i=1}^n I_{y^{(i)} \neq h(\mathbf{x}^{(i)})}.$$

Step 2 — Substitute $h(\mathbf{x}) = \theta$ (constant model):

$$\mathcal{R}_{\text{emp}}(\theta) = \sum_{i=1}^n I_{y^{(i)} \neq \theta}.$$

Intuition:

- ▶ $\mathcal{R}_{\text{emp}}(\theta) = \#$ training points whose label $\neq \theta$.
- ▶ To minimize errors, pick $\theta =$ most frequent $y^{(i)}$, i.e. the **mode**.

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Solution (a) — Instantiating the intuition

Formalize “pick the most frequent $y^{(i)}$ ” as an argmin chain:

$$\arg \min_{\theta \in \mathcal{Y}} \mathcal{R}_{\text{emp}}(\theta) = \arg \min_{\theta \in \mathcal{Y}} \sum_{i=1}^n I_{y^{(i)} \neq \theta}$$

Therefore $\hat{h}(\mathbf{x}) = \text{mode}\{y^{(i)}\}$ is the empirical risk minimizer.

Toy example: $n = 5$ points, labels $\{1, 1, 1, 0, 0\}$.

data $y^{(i)}$: 

Green = hit, red = miss.

$2 < 3$, so $\theta = 1 = \text{mode}\{y^{(i)}\}$ wins.

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Question (b)

Question (b). What is the optimal constant model in terms of the *theoretical* risk for the 0-1-loss, and what is its risk?

Recall. Constant model:

$$\mathcal{H} = \{h \mid h(\mathbf{x}) = \theta \in \mathcal{Y} \forall \mathbf{x} \in \mathcal{X}\}.$$

0-1-loss: $L(y, h(\mathbf{x})) = \mathbf{1}_{y \neq h(\mathbf{x})}$.

Roadmap (same recipe, now for *theoretical* risk)

1. Write down the theoretical risk $\mathcal{R}_L(h)$.
2. Substitute $h(\mathbf{x}) = \theta$ (hypothesis is constant).
3. Marginalize out $\mathbf{x}$ (integrand no longer depends on $\mathbf{x}$).
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Solution (b) — Steps 1–3: write $\mathcal{R}_L$, substitute, marginalize out $\mathbf{x}$

$$\mathcal{R}_L(h) = \int_y \int_{\mathbf{x}} I_{y \neq h(\mathbf{x})} p(\mathbf{x}, y) \, d\mathbf{x} \, dy$$

(Step 1: write the theoretical risk)

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$$\begin{aligned}\mathcal{R}_L(h) &= \int_y \int_{\mathbf{x}} I_{y \neq h(\mathbf{x})} p(\mathbf{x}, y) \, d\mathbf{x} \, dy && \text{(Step 1: write the theoretical risk)} \\ &= \int_y \int_{\mathbf{x}} I_{y \neq \theta} p(\mathbf{x}, y) \, d\mathbf{x} \, dy && \text{(Step 2: substitute } h(\mathbf{x}) = \theta\text{)}\end{aligned}$$

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$$= \int_y \int_{\mathbf{x}} I_{y \neq \theta} p(\mathbf{x}, y) d\mathbf{x} dy$$

(Step 2: substitute $h(\mathbf{x}) = \theta$)

$$= \int_y I_{y \neq \theta} p(y) dy$$

(Step 3: integrand is $\mathbf{x}$ -free, marginalize)

Solution (b) — Step 4: minimize in θ

From Step 3:

$$\mathcal{R}_L(\theta) = \int_{\mathcal{Y}} I_{y \neq \theta} p(y) \, dy = \mathbb{E}[I_{y \neq \theta}] = \mathbb{P}(y \neq \theta), \quad \mathbb{E}[I_A] = \mathbb{P}(A).$$

Use the complement:

$$\mathcal{R}_L(\theta) = \mathbb{P}(y \neq \theta) = 1 - \mathbb{P}(y = \theta).$$

Minimizing risk therefore means choosing the most likely label:

$$\begin{aligned} \arg \min_{\theta \in \mathcal{Y}} \mathcal{R}_L(\theta) &= \arg \min_{\theta \in \mathcal{Y}} (1 - \mathbb{P}(y = \theta)) \\ &= \arg \max_{l \in \mathcal{Y}} \mathbb{P}(y = l). \end{aligned}$$

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From Step 4, the optimal constant model is

$$\bar{h}(\mathbf{x}) = \arg \max_{l \in \mathcal{Y}} \mathbb{P}(y = l),$$

i.e. the label that attains the maximum of $\mathbb{P}(y = l)$, so

$$\mathbb{P}(y = \bar{h}(\mathbf{x})) = \max_{l \in \mathcal{Y}} \mathbb{P}(y = l).$$

Plug into $\mathcal{R}_L(\theta) = 1 - \mathbb{P}(y = \theta)$ (from Step 4):

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Summary of (b) — General takeaways

- **ERM and theoretical risk minimization share one recipe** (for constant models) — write, substitute, simplify, minimize. The only difference is how you average:

$$\underbrace{\frac{1}{n} \sum_{i=1}^n L(y^{(i)}, h(\mathbf{x}^{(i)}))}_{\mathcal{R}_{\text{emp}}(h) \text{ (sample)}} \longleftrightarrow \underbrace{\mathbb{E}_{\mathbf{x}, y}[L(y, h(\mathbf{x}))]}_{\mathcal{R}_L(h) \text{ (population)}}.$$

Here that yields $\text{mode}\{y^{(i)}\} \longleftrightarrow \text{mode of } p(y)$.

- **The loss selects the statistic** you end up estimating.
0-1 loss $\Rightarrow$ *mode*; L2 loss $\Rightarrow$ *mean*; L1 loss $\Rightarrow$ *median*.
- **Indicator trick:** $\mathbb{E}[I_A] = \mathbb{P}(A)$. Turns 0-1-loss integrals into probabilities — keeps the algebra clean and makes the answer interpretable.
- **Hypothesis class $\mathcal{H}$ controls what you can exploit.**
Constant models drop $\mathbf{x} \Rightarrow$ only the *marginal* $p(y)$ matters. A richer $\mathcal{H}$ that depends on $\mathbf{x}$ would let you target $p(y \mid \mathbf{x})$ pointwise (Bayes classifier).

Question (c)

Question (c). Derive the approximation error if the hypothesis space $\mathcal{H}$ consists of the *constant models*.

1. How is the approximation error defined?

$$\inf_{h \in \mathcal{H}} \mathcal{R}_L(h) - \mathcal{R}_L^*$$

2. What do the two terms mean?

$\inf_{h \in \mathcal{H}} \mathcal{R}_L(h)$: best risk *inside* $\mathcal{H}$. $\mathcal{R}_L^*$: best risk over *all* h (Bayes-optimal).

3. Which $\mathcal{H}$ are we using here?

Constant models — same setup as (a) and (b).

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Question (c) — Using (a) and (b) for both terms

First term (from (b)):

$$\inf_{h \in \mathcal{H}} \mathcal{R}_L(h) = 1 - \max_{l \in \mathcal{Y}} \mathbb{P}(y = l).$$

Second term — tower property:

$$\mathcal{R}_L^* = \mathbb{E}_{\mathbf{x}, y} [I_{y \neq h^*(\mathbf{x})}] = \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [I_{y \neq h^*(\mathbf{x})} | \mathbf{x}]].$$

Minimize the inner expectation pointwise — this is (b) with $p(y | \mathbf{x})$ in place of $p(y)$:

$$\mathbb{E}_{y|\mathbf{x}} [I_{y \neq h^*(\mathbf{x})} | \mathbf{x}] = 1 - \max_{l \in \mathcal{Y}} \mathbb{P}(y = l | \mathbf{x}).$$

Take expectation over $\mathbf{x}$:

$$\mathcal{R}_L^* = 1 - \mathbb{E}_{\mathbf{x}} \left[\max_{l \in \mathcal{Y}} \mathbb{P}(y = l | \mathbf{x}) \right].$$

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Solution to (c)

$$\begin{aligned}\inf_{h \in \mathcal{H}} \mathcal{R}_L(h) - \mathcal{R}_L^* &= \mathcal{R}_L(\bar{h}) - \mathcal{R}_L^* \\ &= (1 - \max_{l \in \mathcal{Y}} \mathbb{P}(y = l)) - (1 - \mathbb{E}_{\mathbf{x}}[\max_{l \in \mathcal{Y}} \mathbb{P}(y = l | \mathbf{x})]) \\ &= \mathbb{E}_{\mathbf{x}}[\max_{l \in \mathcal{Y}} \mathbb{P}(y = l | \mathbf{x})] - \max_{l \in \mathcal{Y}} \mathbb{P}(y = l).\end{aligned}$$

Question (d)

(d) Assume now $g = 2$ (binary classification) and consider now the hypothesis space of probabilistic classifiers $\mathcal{H} = \{\pi : \mathcal{X} \rightarrow [0, 1]\}$, that is, $\pi(\mathbf{x})$ (or $1 - \pi(\mathbf{x})$) is an estimate of the posterior distribution $p_{y|\mathbf{x}}(1|\mathbf{x})$ (or $p_{y|\mathbf{x}}(0|\mathbf{x})$). Furthermore, consider the probabilistic 0-1-loss

$$L(y, \pi(\mathbf{x})) = \begin{cases} 1, & \text{if } (\pi(\mathbf{x}) \geq 1/2 \text{ and } y = 0) \text{ or } (\pi(\mathbf{x}) < 1/2 \text{ and } y = 1), \\ 0, & \text{else.} \end{cases}$$

- (i) Is the minimum of $\mathbb{E}_{xy}[L(y, \pi(\mathbf{x}))]$ unique over $\pi \in \mathcal{H}$?
- (ii) Is the posterior distribution $p_{y|\mathbf{x}}$ a resp. the minimizer of $\mathbb{E}_{xy}[L(y, \pi(\mathbf{x}))]$? Discuss the corresponding (dis-)advantages of your findings.

Solution to Question (d) — Setup

Step 1: Indicator form.

	$y = 0$	$y = 1$
$\pi(\mathbf{x}) \geq 1/2$	1	0
$\pi(\mathbf{x}) < 1/2$	0	1

Sum the two “ $L = 1$ ” cells:

$$L(y, \pi(\mathbf{x})) = \mathbb{I}_{\pi(\mathbf{x}) \geq 1/2} \mathbb{I}_{y=0} + \mathbb{I}_{\pi(\mathbf{x}) < 1/2} \mathbb{I}_{y=1}.$$

Step 2: Pointwise.

$\mathcal{H} = \{\pi : \mathcal{X} \rightarrow [0, 1]\} \Rightarrow$ choose $\pi(\mathbf{x})$ independently at each $\mathbf{x}$.

Step 3: Tower property.

$$\mathbb{E}_{xy}[L] = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[L \mid \mathbf{x}]].$$

Minimize $\mathbb{E}_{y|\mathbf{x}}[L \mid \mathbf{x}]$ pointwise.

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Sum the two “ $L = 1$ ” cells:

$$L(y, \pi(\mathbf{x})) = \mathbb{I}_{\pi(\mathbf{x}) \geq 1/2} \mathbb{I}_{y=0} + \mathbb{I}_{\pi(\mathbf{x}) < 1/2} \mathbb{I}_{y=1}.$$

Step 2: Pointwise.

$\mathcal{H} = \{\pi : \mathcal{X} \rightarrow [0, 1]\} \Rightarrow$ choose $\pi(\mathbf{x})$ independently at each $\mathbf{x}$.

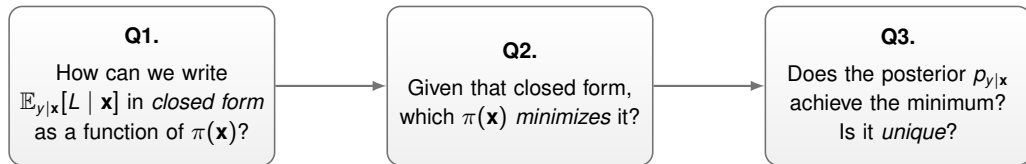
Step 3: Tower property.

$$\mathbb{E}_{xy}[L] = \mathbb{E}_{\mathbf{x}}[\mathbb{E}_{y|\mathbf{x}}[L \mid \mathbf{x}]].$$

Minimize $\mathbb{E}_{y|\mathbf{x}}[L \mid \mathbf{x}]$ pointwise.

Solution to Question (d) — Roadmap

From Step 3 we need to minimize $\mathbb{E}_{y|\mathbf{x}}[L \mid \mathbf{x}]$ at each $\mathbf{x}$. Think through three guiding questions:



Solution to Question (d) — Compute $\mathbb{E}_{y|\mathbf{x}}[L \mid \mathbf{x}]$

$$\begin{aligned}\mathbb{E}_{y|\mathbf{x}}[L(y, \pi(\mathbf{x})) \mid \mathbf{x}] \\ = \mathbb{E}_{y|\mathbf{x}}[I_{\pi(\mathbf{x}) \geq 1/2} I_{y=0} + I_{\pi(\mathbf{x}) < 1/2} I_{y=1} \mid \mathbf{x}]\end{aligned}$$

Why $\mathbb{E}_{y|\mathbf{x}}[I_{y=l} \mid \mathbf{x}] = \mathbb{P}(y = l \mid \mathbf{x})$?

An indicator only takes values $\{0, 1\}$:

$$I_{y=l} = \begin{cases} 1, & y = l, \\ 0, & y \neq l. \end{cases}$$

So its conditional expectation is

$$\begin{aligned}\mathbb{E}_{y|\mathbf{x}}[I_{y=l} \mid \mathbf{x}] &= 1 \cdot \mathbb{P}(y = l \mid \mathbf{x}) \\ &\quad + 0 \cdot \mathbb{P}(y \neq l \mid \mathbf{x}) \\ &= \mathbb{P}(y = l \mid \mathbf{x}).\end{aligned}$$

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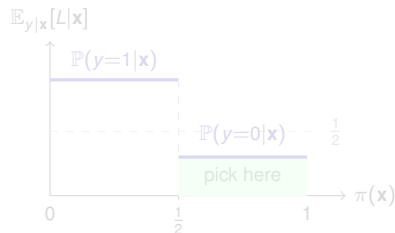
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Case 1. $\mathbb{P}(y = 0 | \mathbf{x}) \geq 1/2$



$$\Rightarrow \pi(\mathbf{x}) < 1/2, \mathbb{E}_{y|\mathbf{x}}[L | \mathbf{x}] = \mathbb{P}(y=1 | \mathbf{x})$$

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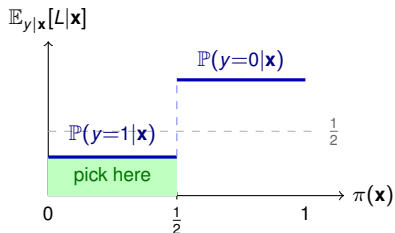
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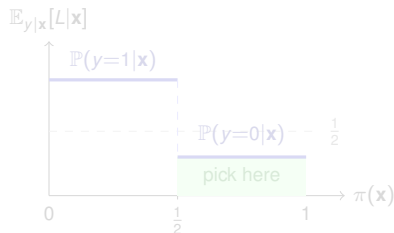
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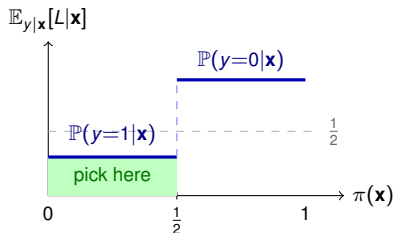
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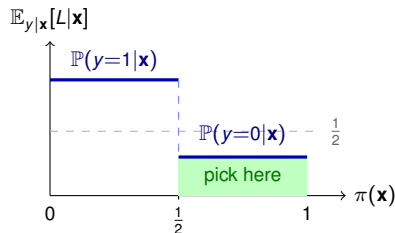
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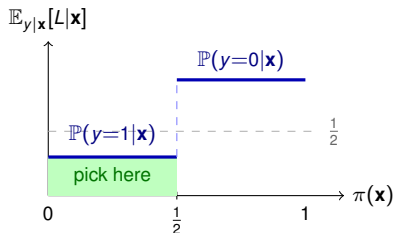
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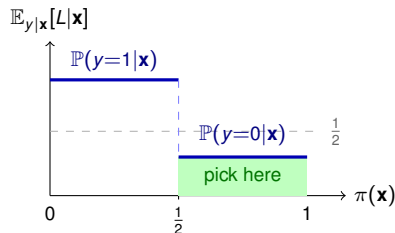
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Solution to Question (d) — Minimizer & (non-)uniqueness

Combining both cases, any minimizer satisfies

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The posterior $p_{y|\mathbf{x}}(1 \mid \mathbf{x})$ is naturally of this form — but it is *not* the only π of this kind: any choice on the correct side of $1/2$ works. **Conclusion (i):** the minimizer is **not unique**.

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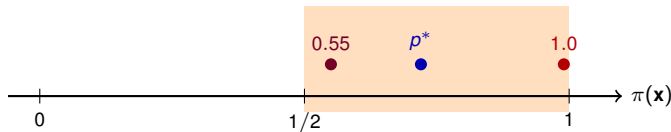
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Using probabilistic 0-1-loss to train a probabilistic classifier; let $p^* := p(y=1 \mid \mathbf{x})$ and assume $p^* > 1/2$:

All shaded $\pi(\mathbf{x})$ are optimal — same loss:



Posterior p^* and extremes (e.g. 1.0) tie.

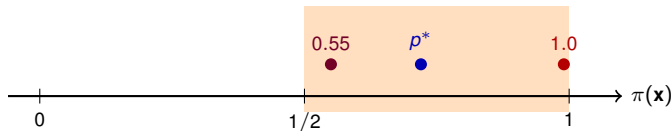
Pro. The optimum lands on the correct side of $1/2 \Rightarrow$ thresholding at $1/2$ predicts the more likely class given $\mathbf{x}$, i.e. $\arg \max_l p(y=l \mid \mathbf{x})$.

Con. All values on that side tie ($\pi(\mathbf{x}) = 1.0$ is as “optimal” as $\pi(\mathbf{x}) = p^*$) $\Rightarrow$ minimizing it does not recover the actual *probability* p^* .

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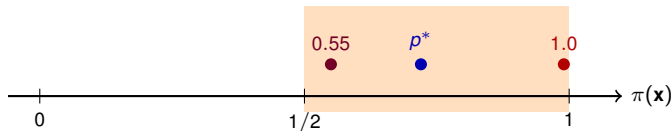
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Summary — Connecting the exercise to the lecture

What we derived	Lecture concept made concrete
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(c) $\inf_{\mathcal{H}} \mathcal{R}_L - \mathcal{R}_L^* = \mathbb{E}_{\mathbf{x}}[\max_l p(y=l \mathbf{x})] - \max_l p(y=l)$	Bayes risk + approximation error decomposition
(d) Posterior is a minimizer, but $\pi(\mathbf{x}) \equiv 1$ is too	0-1 is <i>proper</i> but not strictly proper

Take-aways

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